INSTRUCTIONS

PRELIMINARY GEK-45144

MOTOR FIELD REVERSING CARD

531X168FRGACG1

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

GENERAL ELECTRIC

FIELD REVERSING CONTACTOR CONTROL

GENERAL

The FRC card contains the logic, level detection and timing circuitry to operate a forward and a reverse field reversing contactor when used in conjunction with the remote motor field control assembly (MFR)

The contactor coils are operated at 105V DC (115V AC full wave rectified, unfiltered supply). The contactor coils are controlled directly from the FRC card by two photo isolated transistor switches. Maximum switch (coil) rating is 350mA. The input power supply is +15V, 75mA maximum.

DESCRIPTION OF OPERATION

The sequence of operation when switching from forward to reverse is:

1. A reverse operating command is given by connecting **REV** (tab 3) to common.

   a) If the CEMF (5) voltage is high (motor operating above 20% voltage) or if the CEMF latch is energized by having +15V applied at RUN (4) the only reaction to the reverse command is that the $\overline{\text{ZR}}$ (8X) will be pulled low (+ .6V) (The $\overline{\text{ZR}}$ signal may be used as an option to reduce the drive reference voltage).

   b) If the CEMF latch is off and if the CEMF voltage is reduced below the 5% to 15% level set by the SPD potentiometer a low speed condition is detected making the LSP voltage switch high. This initiates the reversing sequence.

2. When a low speed is detected the RF(7) output switches high to phase back the firing of the MFR field supply, the field current and the FC voltage starts reducing. When the FC voltage is reduced to low level, 5% to 10%, adjustable by the FLD potentiometer a "low field" is detected and the LFD voltage switches high.

3. When a low field is detected the $\overline{\text{FO}}$(8) is pulled low (.6V) to inhibit further firing of the MFR field supply and a time delay, .5 sec. to 3 sec. adjustable by the TD1 potentiometer is initiated. At the end of the TD1 time delay, the forward contactor switch, Q1, is opened and the forward contactor, FC, should drop out.

4. At the end of TD1, a second time delay, adjustable from .5 sec. to 3 sec. by the TD2, potentiometer is initiated. At the end of TD2, the reverse contactor switch, Q2 is closed and the reverse contactor, RC, should pick up, the $\overline{\text{FQ}}$ voltage should switch high to restart the MFR firing and the RF voltage should go low to release the MFR reference.

5. As field current increases in the reverse direction low field detection signal LFD switches low and the $\overline{\text{ZR}}$ voltage goes high (to release the drive reference if used).

6. A contactor fault is indicated when the red FALT light turns on and by the **CFT** voltage switching low. The action takes place at the end of the TD2 time delay stage: If the auxiliary forward contactor interlock has not connected the FCX(10X) input to common to indicate contactor dropout the reverse contactor switch Q2 will not close and only the red FALT light will be on.

NOTE: This fault is not latched in. It is reset if a forward command is given or if the FCX input is connected to common.

7. Switching from reverse to forward operation follows the same sequence as listed above except the Q2 switch opens and the RCX (9X) input must be connected to common by the opening of the RC contactor before the Q1 switch closes.

FAULT

A fault condition is not detected if a contactor fails to pick up.

1. A fault condition may be the result of one (or both) of the transistor switches, Q1 or Q2, shorting out to keep a contactor picked up.

2. A failure in the card logic circuitry could also cause a fault.

3. If the contactor does drop out, the auxiliary interlock may fail to connect its respective input to common.

FIELD REVERSING CONTROL

INPUTS NOM.	TAB		FROM/TO	DESCRIPTION
$\overline{\text{REV}}$	3		EXT	FWD/REV command: +15V or N.C. - FWD; zero V or connected to common - REV.
CEMF	5		MFR	Motor armature voltage (CEMF or Tach) feedback. (4 volts at rated armature volts)
RUN	4		EXT	CEMF latch (optional, to inhibit low speed detection if +15V)
FC	6		MFR	Motor field current feedback. (2V to 4V at rated field current)
+15V	2	2X	EXT	+15V regulated power supply.
COM	1	1X	EXT	Signal common.
FCX	10X		FC	Fwd. contactor auxiliary contacts
RCX	9X		RC	Rev. contactor auxiliary contacts
OUTPUTS				
CFT	7X		EXT	Contactor Fault Trip output. (Contactor interlock not closing following a dropout command.)
$\overline{\text{ZR}}$	8X		EXT	Reference clamp. For automatic reversing by REV command.
RF	7		MFR	Reduce field reference command.
$\overline{\text{FO}}$	8		MFR	Zero field (stop firing) command.
FWDP	20	20X	FC	Positive transistor switch output.
FWDN	23	23X	FC	Negative transistor switch output.
REVP	29	29X	RC	Positive transistor switch output.
REVN	26	26X	RC	Negative transistor switch output.

ADJUSTMENTS

SPD - Low Speed (CEMF) detection, CCW - 5%, CW - 15%

FCAL - Adjust SFC = 10.0 volts with rated FC voltage.

FLD - Low field (FC) detection, CCW - 5%, CW - 10%

TD1 - Time delay contactor dropout, CCW - .5 sec., CW - 3 sec.

TD2 - Time delay contactor pickup. CCW - .5 sec., CW - 3 sec.

OTHER NOMENCLATURE

SFC - Scaled FC voltage test point (SFC = FC with FCAL set CCW. Adjust SFC = 10V with FC at rated voltage)

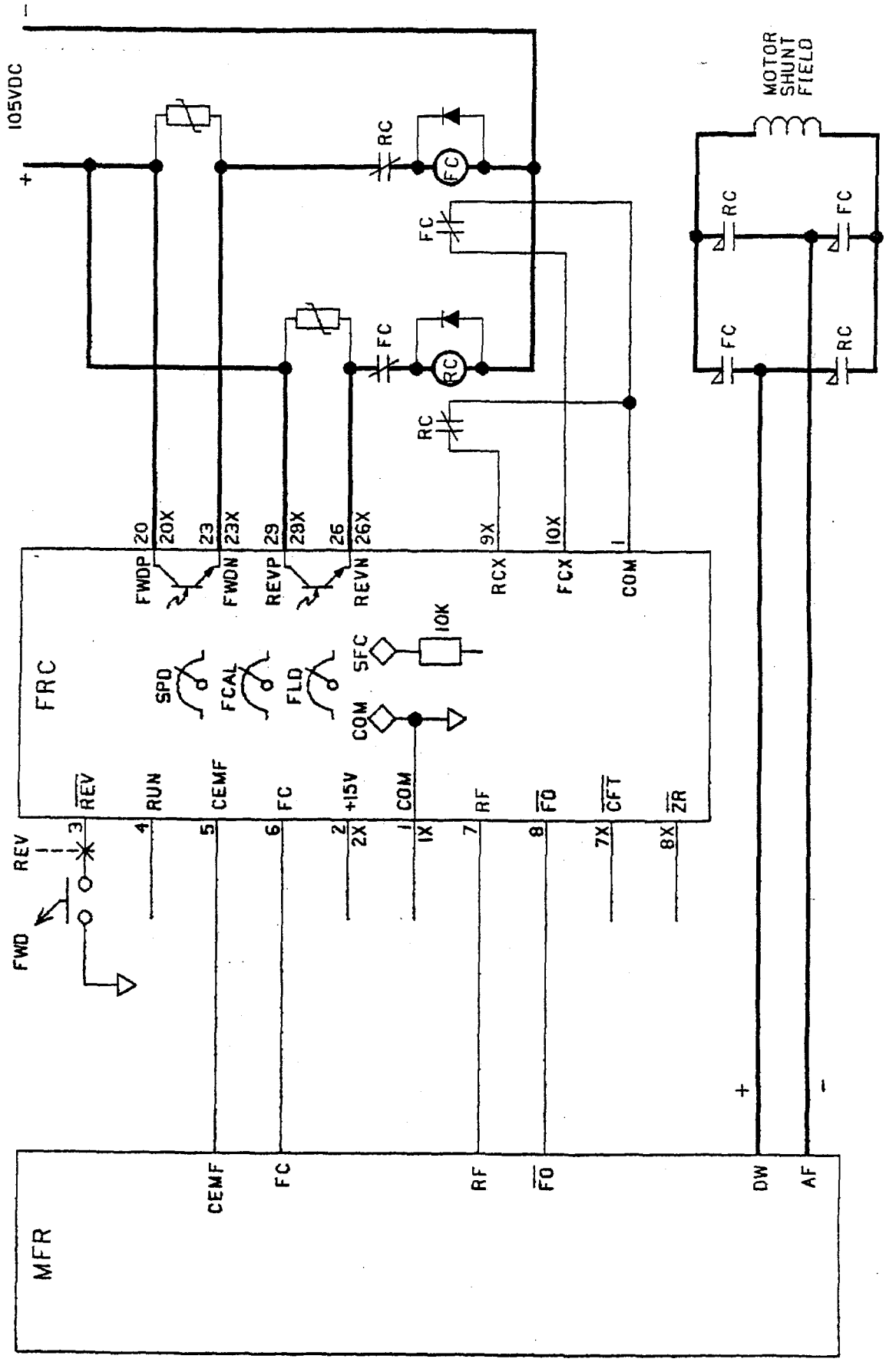
FC - Fwd. contactor

RC - Rev. contactor

MFR - Motor Field controlled voltage supply

FRC - Field Reversing Contactor control card

EXT - External connection



FIELD REVERSING CONTROL
CONNECTION DIAGRAM

INITIAL SET-UP AND ADJUSTMENTS

CAUTION

INCORRECT WIRING MAY RESULT IN MISOPERATION. ACCIDENTAL SHORTING OF THE FIELD SUPPLY MAY CAUSE FUSE AND COMPONENT FAILURES.

1. Bypass the FC and RC contactors by connecting the field directly to the MFR output DW and AF. Also, disconnect the main FC and RC contacts from the MFR and the field. In this mode the MFR and the FRC can be adjusted and proper contactor operation checked out prior to actual reversal of the field.

2. Wire check the connections between the FRC card and FC and RC contactor coils and auxiliary interlocks.

NOTE: The static switches (transistors) on the FRC card are unidirectional. The positive side of the 104V DC coil supply must connect to the positive switch terminals, FWDP and REVP.

   Verify that the FC and RC coils each have a free-wheeling diode connected across the coils and make sure the cathode points toward the positive side of the coil when energized.

3. Preliminary adjustments of potentiometers.

SPD (P2) - 50% Initiate field reduction at or below - 10% voltage

FACL (P1) - CCW SFC volts = FC volts

FLD (P3) - 50% Start reversing sequence at about - 7% field current

TD1 (P4) - 50% ≈ 2 sec. delay to drop contactor

TD2 (P5) - 50% ≈ 2 sec. delay to pick up contactor

   After the above and other preliminary checkouts have been completed power can be applied to the drive.

4. Assume FWD operation is selected and power applied:

   a) Initially no lights on FRC card should indicate and the field voltage should stay low.

   b) After a time delay of about 2 seconds the FWD light should come on and the FC contactor should pick up (if the 105V DC also has been applied). Also, at this time $\overline{\text{FO}}$ should go high (+15V) to initiate MFR firing and RF should go low to release the MFR reference. Field voltage should now be applied and the field current should build up to normal level.

NOTE: The field current may now be less than rated if field economy is programmed in.

5. With FCAL set CCW, the voltage at testpoint SFC=FC. If field economy is applied a drive START must be initiated in order for the field current to reach rated level. With the field current at rated level (FC=2 to 4 volts) adjust the FCAL pot CW until SFC=10±.1 volt. This scales the FC feedback such that "low field detection" set by FLD covers a range of 5% (CCW) to 10% (CW) of rated field current.

6. Operate the drive at a speed exceeding 20% rated voltage (or apply +15 volts at RUN). Initiate a REV command by connecting $\overline{\text{REV}}$ to common. The voltage at $\overline{\text{ZR}}$ should go low, but no other change should occur (unless $\overline{\text{ZR}}$ voltage is used to clamp the drive reference for automatic reversal). Reset the directional command to FWD and stop the drive (or remove +15V from RUN).

7. With the drive stopped, initiate a REV command. RF should switch high and field current should start reducing. At about 7-8% field current a "low field" detection should occur, $\overline{\text{FO}}$ should go low and a time delay is initiated. After about 2 seconds (TD1 at 50%) the FWD light should go out and the FC contactor should drop out. A second time delay is now initiated. After about 2 seconds (TD2 at 50%) the REV light should come on and the RC contactor should pick up. The $\overline{\text{FO}}$ voltage again goes high, the RF voltage goes low, the field current should increase and th $\overline{\text{ZR}}$ voltage should go high.

8. Turn SPD and FLD fully CCW for minimum voltage and field current requirement and set TD1 and TD2 CW for maximum time delays (about 3 seconds). Repeat step 6 for both REV and FWD operation.

9. After proper field reduction and contactor operation has been established, disconnect drive power and reconnect the FC and RC contactor according to the elementary diagram.

10. Repeat the reversing exercise as in step 6.

NOTE. It is essential that all contactor arcing has stopped prior to picking up the opposite contactor.

11. If the voltage is tailing out too long before a "low speed" is detected turn SPD in the CW direction. Likewise, if the field current is tailing out too long before a "low field" is detected, turn FLD in the CW direction. To speed up the contactor operation the time delays may be reduced by turning TD1 and TD2 in the CCW direction. In general, TD1 and TD2 can be set for the same time delay. Do not set the time delays faster than 1 second (TD1 and TD2 at about 6 O'clock) unless careful monitoring of field current and contactor arcing has been performed.

NOTE: With a free-wheeling diode across the contactor coil the dropout time will be slow. This will effectively add to the TD1 time delay and reduce the TD2 time delay. Do not reduce TD2 such that a pick-up command is given before the opposite contactor has actually dropped out.

12. A fault condition is indicated by the FALT light and the inability to pick up the selected contactor.

    A fault is detected if the FC (RC) auxiliary interlock does not connect FCX (RCX) to common when a command to pick up RC (FC) is attempted.

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